MSG No. 89.93 P_C422 [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `422`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `422`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `4'22'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,5,6,10,11,15,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,7,8,9,12,13,14]`

Table 4: Wyckoff site: `2d`, site symmetry: `4'22'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,5,6,10,11,15,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,7,8,9,12,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,7,14,16]`
2	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[2,5,11,12]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[3,4,10,13]`
4	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[6,8,9,15]`

Table 6: Wyckoff site: `4f`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,7,14,16]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,5,11,12]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,4,10,13]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,8,9,15]`

Table 7: Wyckoff site: `4g`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,6]`
2	$[0,\ 0,\ -z]$	`[4,5,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[9,10,11,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[12,13,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `4'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,10,11]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,9,14]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4,5,15,16]`
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[7,8,12,13]`

Table 9: Wyckoff site: `8i`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,14]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[2,11]`
3	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[3,10]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[4,13]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[5,12]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[6,9]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[7,16]`
8	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[8,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `.2.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,5]`
2	$[-y,\ 0,\ 0]$	`[2,8]`
3	$[y,\ 0,\ 0]$	`[3,7]`
4	$[0,\ -y,\ 0]$	`[4,6]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[9,13]`
6	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ 0\right]$	`[10,16]`
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[11,15]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[12,14]`

Table 11: Wyckoff site: `8k`, site symmetry: `.2.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,5]`
2	$\left[-y,\ 0,\ \frac{1}{2}\right]$	`[2,8]`

continued ...

Table 11

No.	position	mapping
3	$\left[y,\, 0,\, \frac{1}{2}\right]$	[3,7]
4	$\left[0,\, -y,\, \frac{1}{2}\right]$	[4,6]
5	$\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[9,13]
6	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, \frac{1}{2}\right]$	[10,16]
7	$\left[y+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	[11,15]
8	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[12,14]

Table 12: Wyckoff site: 8l, site symmetry: ..2

No.	position	mapping
1	$[x,\, x,\, 0]$	[1,7]
2	$[-x,\, x,\, 0]$	[2,5]
3	$[x,\, -x,\, 0]$	[3,4]
4	$[-x,\, -x,\, 0]$	[6,8]
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	[9,15]
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, 0\right]$	[10,13]
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$	[11,12]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, 0\right]$	[14,16]

Table 13: Wyckoff site: 8m, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[1,15]
2	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	[2,13]
3	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[3,12]
4	$\left[x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[4,11]
5	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[5,10]
6	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[6,16]
7	$\left[x+\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[7,9]
8	$\left[\frac{1}{2}-x,\, -x,\, \frac{1}{2}\right]$	[8,14]

Table 14: Wyckoff site: 8n, site symmetry: ..2

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{2}\right]$	[1,7]
2	$\left[-x,\, x,\, \frac{1}{2}\right]$	[2,5]
3	$\left[x,\, -x,\, \frac{1}{2}\right]$	[3,4]
4	$\left[-x,\, -x,\, \frac{1}{2}\right]$	[6,8]
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[9,15]
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[10,13]
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[11,12]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[14,16]

Table 15: Wyckoff site: 8o, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	[1,15]
2	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	[2,13]
3	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	[3,12]
4	$\left[x,\, \frac{1}{2}-x,\, 0\right]$	[4,11]
5	$\left[-x,\, x+\frac{1}{2},\, 0\right]$	[5,10]
6	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	[6,16]
7	$\left[x+\frac{1}{2},\, x,\, 0\right]$	[7,9]
8	$\left[\frac{1}{2}-x,\, -x,\, 0\right]$	[8,14]

Table 16: Wyckoff site: 16p, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-y,\, x,\, z]$	[2]
3	$[y,\, -x,\, z]$	[3]
4	$[x,\, -y,\, -z]$	[4]
5	$[-x,\, y,\, -z]$	[5]
6	$[-x,\, -y,\, z]$	[6]
7	$[y,\, x,\, -z]$	[7]
8	$[-y,\, -x,\, -z]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[9]
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	[10]
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	[11]
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[12]
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[13]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[14]
15	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[15]
16	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[16]